

Numerical Analysis 1, Final Exam, 3.Session

a) Derive a numerical integration formula for $\int_0^1 f(x) dx$ based on the Lagrange interpolation polynomial at the nodes -2, -1, and 0. Apply this result to evaluate the integral when $f(x) = e^x$

b) Evaluate the integral $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos x dx$ by using the Trapezoidal rule with 5 sample points.

c) Determine the number M and the interval width h so that the Trapezoidal rule for M subintervals can be used to compute the integral $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \cos x dx$ with an accuracy of 5×10^{-9} .

Note: Use 3 digits after decimal point.

Duration 40 min